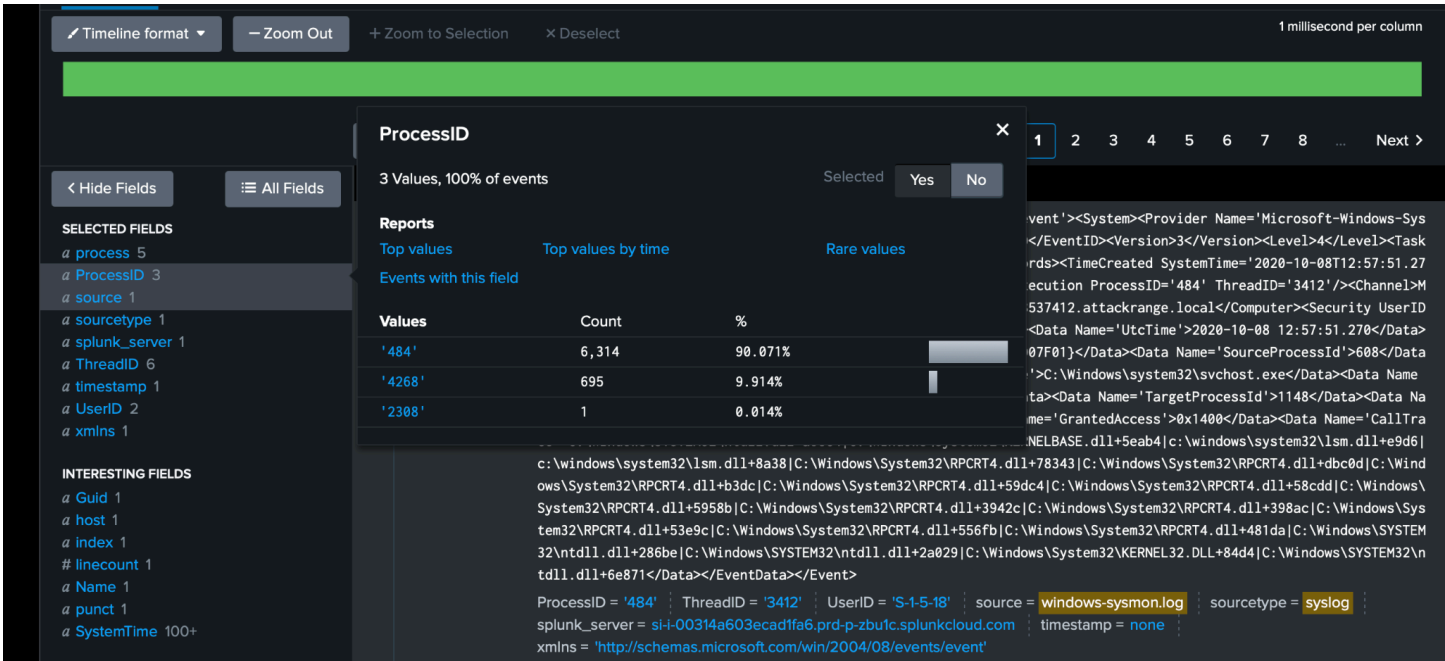
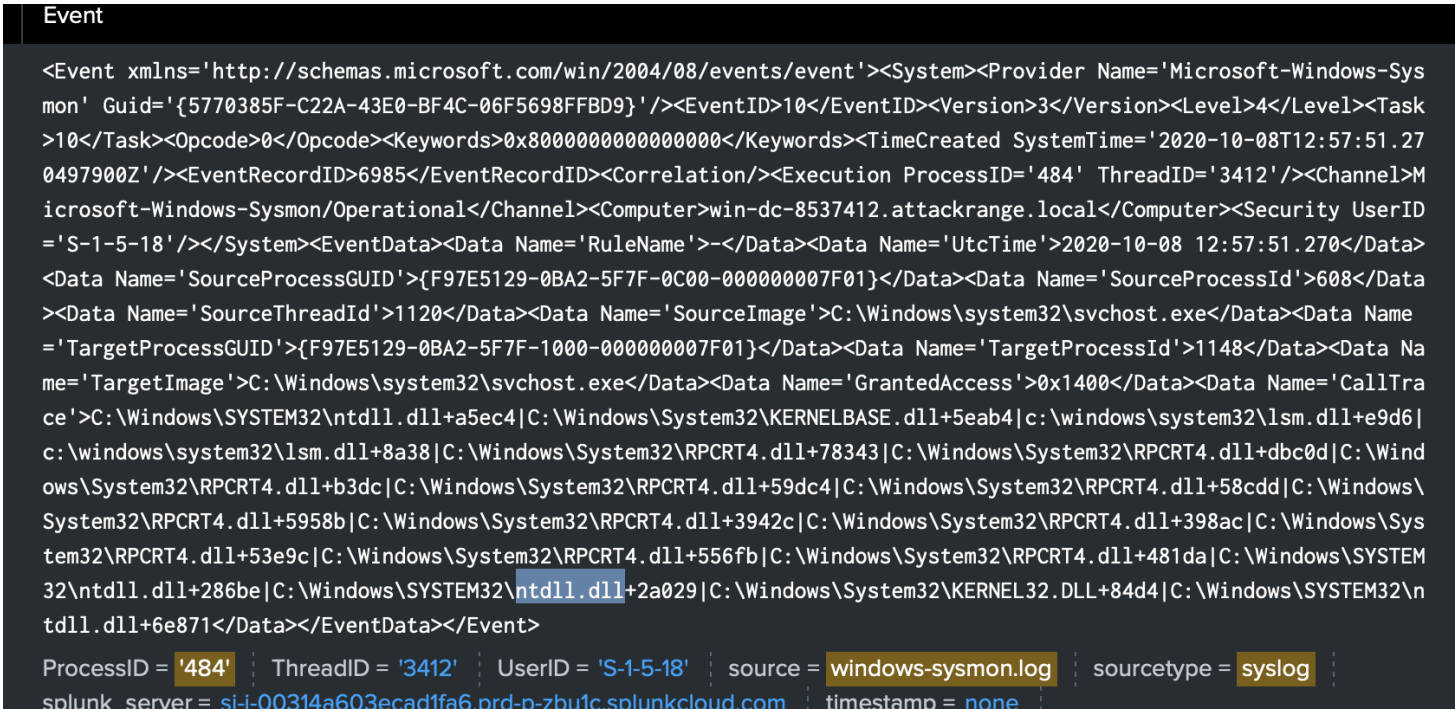


We see that ProcessID 482 is 90.7% of the logs



We see that **ProcessID 484** accounts for approximately **90.7% of the logs**, making it the first process selected for investigation.



We see that almost all events associated with this process are **Sysmon Event ID 10**.

Using the following query:

```
source="windows-sysmon.log" host="si-i-00314a603ecad1fa6.prd-p-zbulc.splunkcloud.com" sourcetype="syslog"
ProcessID="484"
| rex field=_raw "<Task>(?!<TaskID>[^\<]+)</Task>"
| stats count by TaskID
| sort - count
```

Sysmon Event ID 10 = Process Access, which means one process attempted to access another process. This event type is commonly used to detect process injection, credential dumping, and suspicious memory access activity.

TaskID ↕	count ▼
10	5647
1	317
11	199
22	77
13	32
3	25
12	8
6	8
4	1

During analysis of the Sysmon logs, multiple suspicious Process Access (Event ID 10) events were identified involving **PowerShell**, **svchost.exe**, and **csc.exe**. The activity included **full-access permissions (0x1ffff)** between processes, **PowerShell-to-PowerShell memory access**, and **PowerShell interaction with the .NET C# compiler (csc.exe)**.

These behaviors are consistent with attack preparation activities commonly observed before credential dumping operations.

Suspicious svchost.exe Activity

One event showed:

- SourceImage: svchost.exe
- TargetImage: svchost.exe
- GrantedAccess: 0x1400

This access level includes:

- Query process information
- Read process memory (partial)

This is suspicious because one svchost instance is accessing another svchost process.

Another event showed:

- SourceProcessId: 608
- TargetProcessId: 1148

This mismatch indicates one svchost process was probing another process, which may indicate service abuse or process manipulation.

PowerShell Activity

Another Process ID was associated with PowerShell activity.

When investigating the PowerShell events, one event stood out as strong evidence of suspicious behavior.

The event showed:

?

Q

|

Splunk

New Search

Save As

Create

```
source="windows-sysmon.log" host="si-i-00314a603ecad1fa6.prd-p-zbu1c.splunkcloud.com" sourcetype="syslog" ProcessID="'4268'"
| rex field=_raw "<Data Name='SourceImage'>(?(Commad1>[^\<]+)</Data>"
| table Commad1
| stats count by Commad1
| sort - count
```

695 events (before 6/24/26 1:57:54.000 AM)

No Event Sampling

Job

Policy-Based Pool

Events

Patterns

Statistics (11)

Visualization

Show: 20 Per Page

Format

Preview: On

Commad1

C:\Windows\system32\svchost.exe

C:\Windows\system32\lsass.exe

C:\Windows\system32\csrss.exe

C:\Windows\system32\conhost.exe

C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

C:\Windows\system32\WinrsHost.exe

C:\Windows\system32\cmd.exe

C:\Windows\Microsoft.NET\Framework64\v4.0.30319\csc.exe

C:\Windows\system32\services.exe

C:\Windows\servicing\TrustedInstaller.exe

C:\Windows\system32\LogonUI.exe

- SourceImage: powershell.exe

- TargetImage: powershell.exe

Event	
M	<pre><Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385F-C22A-43E0-BF4C-06F5698FFBD9}' /><EventID>10</EventID><Version>3</Version><Level>4</Level><Task>10</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2020-10-08T12:52:15.742807100Z' /><EventRecordID>516</EventRecordID><Correlation/><Execution ProcessID='4268' ThreadID='4756' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>win-dc-8537412.attackrange.local</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'>-</Data><Data Name='UtcTime'>2020-10-08 12:52:15.727</Data><Data Name='SourceProcessGUID'>{F97E5129-0B7F-5F7F-1303-000000007E01}</Data><Data Name='SourceProcessId'>2776</Data><Data Name='SourceThreadId'>4620</Data><Data Name='SourceImage'>C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe</Data><Data Name='TargetProcessGUID'>{F97E5129-0B7F-5F7F-1403-000000007E01}</Data><Data Name='TargetProcessId'>2724</Data><Data Name='TargetImage'>C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe</Data><Data Name='GrantedAccess'>0x1fffff</Data><Data Name='CallTrace'>C:\Windows\SYSTEM32\ntdll.dll+a7194 C:\Windows\System32\KERNELBASE.dll+2b860 C:\Windows\System32\KERNELBASE.dll+6b246 C:\Windows\System32\KERNEL32.DLL+1c213 C:\Windows\assembly\NativeImages_v4.0.30319_64\System\78b8fb2c58a4cdcc3a44547b9bbd80b9\System.ni.dll+3332f6 C:\Windows\assembly\NativeImages_v4.0.30319_64\System\78b8fb2c58a4cdcc3a44547b9bbd80b9\System.ni.dll+2b5560 C:\Windows\assembly\NativeImages_v4.0.30319_64\System\78b8fb2c58a4cdcc3a44547b9bbd80b9\System.ni.dll+2b4f07 C:\Windows\assembly\NativeImages_v4.0.30319_64\System.Manaa57fc8cc#\bbcfce4d5e2ff289fc26db1642aedc89\System.Management.Automation.ni.dll+b5a321b(wow64) C:\Windows\assembly\NativeImages_v4.0.30319_64\System.Manaa57fc8cc#\bbcfce4d5e2ff289fc26db1642aedc89\System.Management.Automation.ni.dll+aa440a5(wow64) C:\Windows\assembly\NativeImages_v4.0.30319_64\System.Manaa57fc8cc#\bbcfce4d5e2ff289fc26db1642aedc89\System.Management.Automation.ni.dll+aa43d76(wow64) C:\Windows\assembly\NativeImages_v4.0.30319_64\System.Manaa57fc8cc#\bbcfce4d5e2ff289fc26db1642aedc89\System.Management.Automation.ni.dll+b4f53db(wow64) C:\Windows\assembly\NativeImages_v4.0.30319_64\System.Manaa57fc8cc#\bbcfce4d5e2ff289fc26db1642aedc89\System.Management.Automation.ni.dll+aa0490c(wow64) C:\Windows\assembly\NativeImages_v4.0.30319_64\System.Manaa57fc8cc#\bbcfce4d5e2ff289fc26db1642aedc89\System.Management.Automation.ni.dll+aa62ddb(wow64) C:\Windows\assembly\NativeImages_v4.0.30319_64\System.Manaa57fc8cc#\bbcfce4d5e2ff289fc26db1642aedc89\System.Management.Automation.ni.dll+aa46440(wow64) C:\Windows\assembly\NativeImages_v4.0.30319_64\System.Manaa57fc8cc#\bbcfce4d5e2ff289fc26db1642aedc89\System.Management.Automation.ni.dll+aa462d1(wow64) C:\Windows\assembly\NativeImages_v4.0.30319_64\System.Manaa57fc8cc#\bbcfce4d5e2ff289fc26db1642aedc89\System.Management.Automation.ni.dll+aa38256(wow64) C:\Windows\assembly\NativeImages_v4.0.30319_64\System.Manaa57fc8cc#\bbcfce4d5e2ff289fc26db1642aedc89\System.Management.Automation.ni.dll+aa44789(wow64) C:\Windows\assembly\NativeImages_v4.0.30319_64\System.Manaa57fc8cc#\bbcfce4d5e2ff289fc26db1642aedc89\System.Management.Automation.ni.dll+aa4437c(wow64) C:\Windows\assembly\NativeImages_v4.0.30319_64\System.Manaa57fc8cc#\bbcfce4d5e2ff289fc26db1642aedc89\System.Management.Automation.ni.dll+aa440a5(wow64) C:\Windows\assembly\NativeImages_v4.0.30319_64\System.Manaa57fc8cc#\bbcfce4d5e2ff289fc26db1642aedc89\System.Management.Automation.ni.dll+aa43d76(wow64) C:\Windows\assembly\NativeImages_v4.0.30319_64\System.Manaa57fc8cc#\bbcfce4d5e2ff289fc26db1642aedc89\System.Management.Automation.ni.dll+b4f53db(wow64) C:\Windows\assembly\NativeImages_v4.0.30319_64\System.Manaa57fc8cc#\bbcfce4d5e2ff289fc26db1642aedc89\System.Management.Automation.ni.dll+aa2abd7(wow64) C:\Windows\assembly\NativeImages_v4.0.30319_64\System.Manaa57fc8cc#\bbcfce4d5e2ff289fc26db1642aedc89\System.Management.Automation.ni.dll+aa2a1a7(wow64)</Data></EventData></Event></pre>

This means one PowerShell process was accessing another PowerShell process. This behavior should be flagged because attackers commonly use PowerShell to inject scripts or execute code in memory.

Another suspicious field was:

GrantedAccess: 0x1fffff

- FULL CONTROL of the target process
- Read and write memory access

- Suspend and resume threads
- Ability to inject code

This is a high-risk indicator because full process access is frequently associated with process injection, malware execution, and credential dumping preparation activities.

Conclusion

Even though I did not capture the typical T1003.003 activity involving tools such as ntdsutil.exe, vssadmin.exe, or direct access to NTDS.dit, the logs still contain suspicious behavior.

Multiple Sysmon Event ID 10 Process Access events were observed, including PowerShell processes accessing other PowerShell processes and requests for high-level access rights such as GrantedAccess 0x1FFFFFFF. This level of access allows full control of the target process, including reading and writing memory and code injection.